

EMCH 260

SOLID MECHANICS

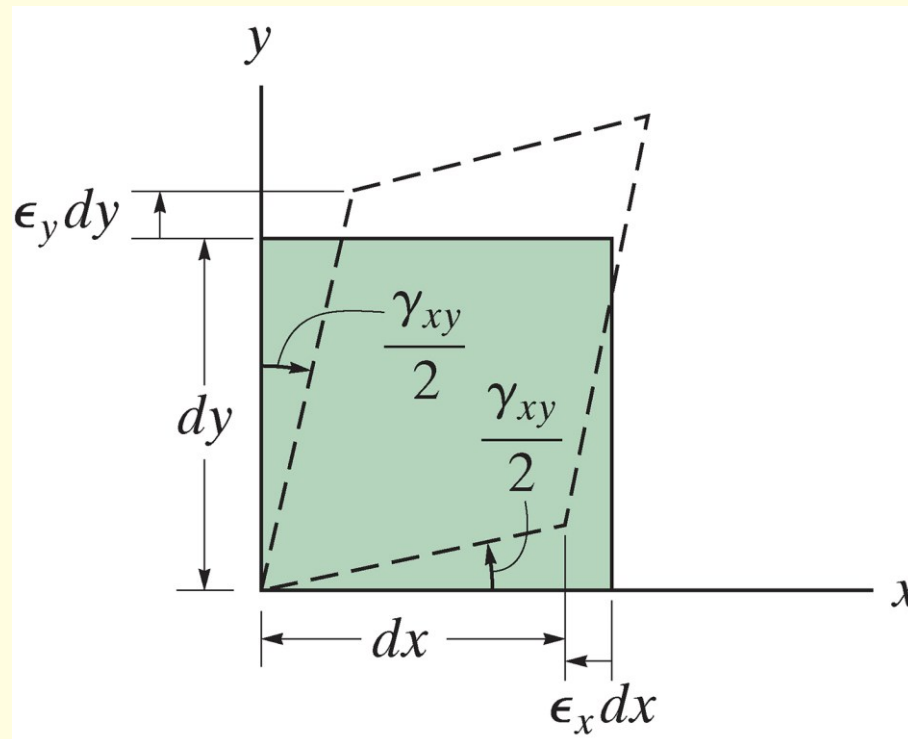
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IN-CLASS EXAMPLE 1

The state of plane strain at a point has strain components of $\epsilon_x = -200 \times 10^{-6}$, $\epsilon_y = 400 \times 10^{-6}$, and $\gamma_{xy} = 120 \times 10^{-6}$.

Determine the maximum in-plane shear strain and the corresponding orientation of the principal plane, and the absolute maximum shear strain at the point.



IN-CLASS EXAMPLE 1, cont'd

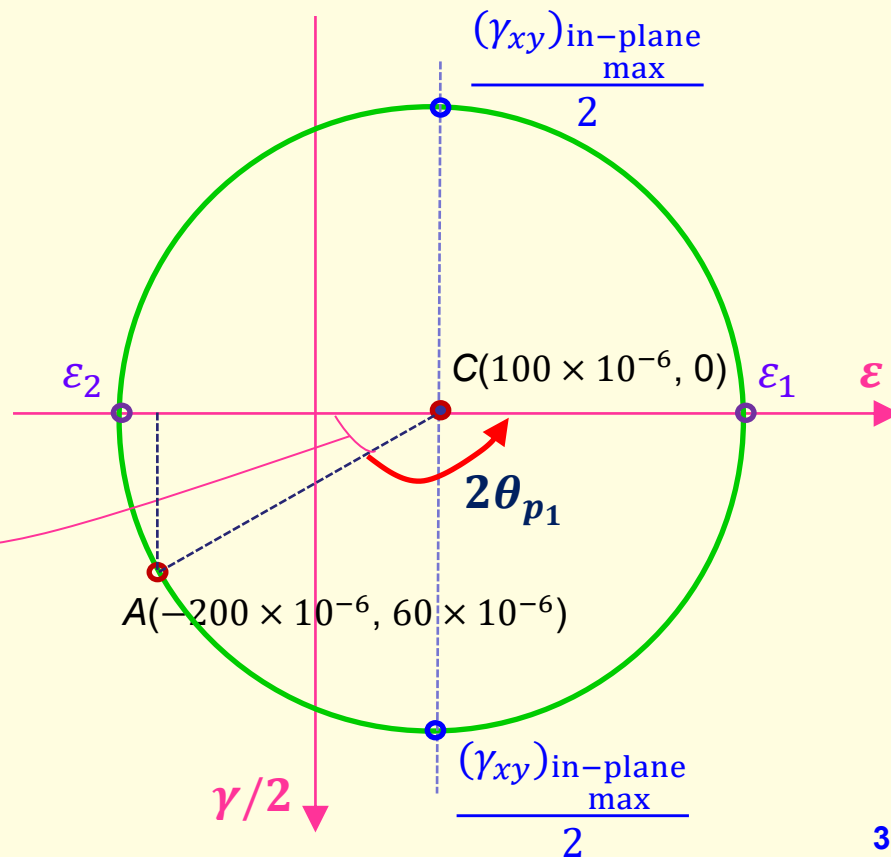
Solutions

Step 1: Construction of the Mohr's Circle of Plane Strain

Given $\varepsilon_x = -200 \times 10^{-6}$, $\varepsilon_y = 400 \times 10^{-6}$, $\gamma_{xy} = 120 \times 10^{-6}$, the center of the Mohr's circle C is located on the ε -axis, at the point $\varepsilon_{\text{avg}} = \frac{\varepsilon_x + \varepsilon_y}{2} = 100 \times 10^{-6}$

The point $C(100 \times 10^{-6}, 0)$ and the reference point $A(-200 \times 10^{-6}, 60 \times 10^{-6})$ are labeled. Then, the radius of the circle is $R = \sqrt{(100 + 200)^2 + 60^2} \times 10^{-6} = 306 \times 10^{-6}$, and the Mohr's circle is constructed (right figure).

$$\tan^{-1} 60/300 = 11.3^\circ$$



IN-CLASS EXAMPLE 1, cont'd

Solutions, Cont'd

Step 2: Principal Strains and Orientation of Principal Plane

$$\varepsilon_1 = (100 + 306) \times 10^{-6} = 406 \times 10^{-6} \quad \textbf{(Ans.)}$$

$$\varepsilon_2 = (100 - 306) \times 10^{-6} = -206 \times 10^{-6} \quad \textbf{(Ans.)}$$

The maximum in-plane shear stress is

$$\gamma_{\max}^{\text{in-plane}} = \varepsilon_1 - \varepsilon_2 = 612 \times 10^{-6} \quad \textbf{(Ans.)}$$

From the circle, the **counterclockwise** angle $2\theta_{p_1}$, measured from CA to the $+\varepsilon$ axis, is

$$2\theta_{p_1} = 180^\circ - \tan^{-1} \left(\frac{60}{300} \right) = 168.7^\circ$$

$$\text{Thus, } \theta_{p_1} = 84.35^\circ \quad \textbf{(Ans.)}$$

The counterclockwise rotation defines the direction of the **x' -axis and ε_1** .

IN-CLASS EXAMPLE 1, cont'd

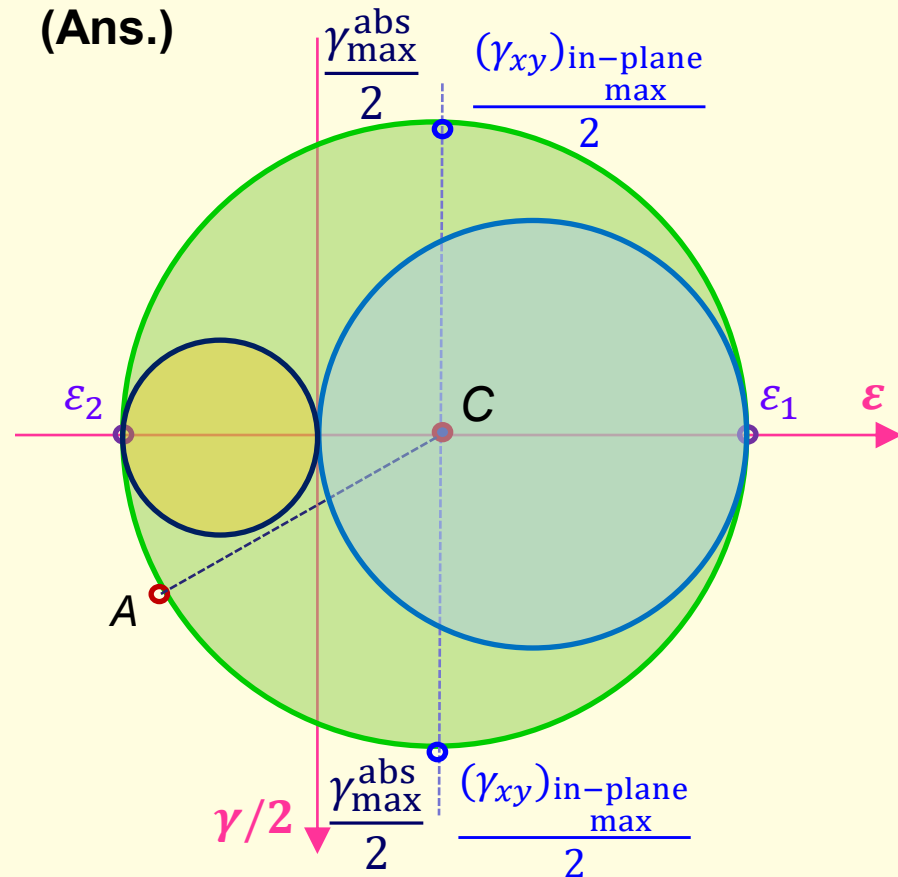
Solutions, Cont'd

Step 3: Absolute Maximum Shear Strain

Since ε_1 and ε_2 have opposite signs, the absolute maximum shear strain is:

$$\gamma_{\max}^{\text{abs}} = \varepsilon_1 - \varepsilon_2 = 612 \times 10^{-6}$$

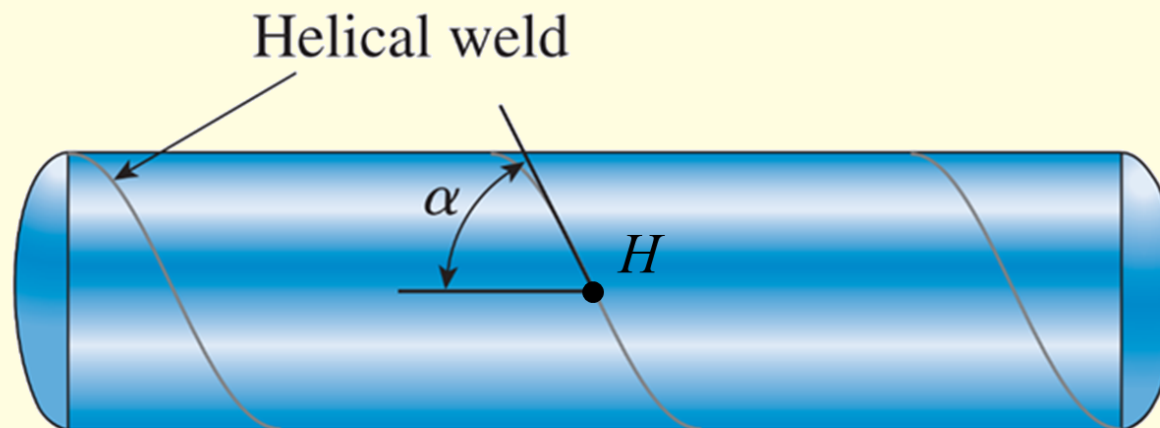
(Ans.)



IN-CLASS EXAMPLE 2

A pressurized steel tank is constructed with a helical weld that makes an angle $\alpha = 55^\circ$ with the longitudinal axis (see figure). The tank has inner radius $r = 0.6$ m, wall thickness $t = 18$ mm, and internal pressure $p = 2.8$ MPa. For a point H on the tank surface, determine the following quantities:

- (a) The circumferential (hoop) and longitudinal stresses.
- (b) The maximum in-plane shear stress and absolute maximum shear stress.
- (c) The normal and shear stresses acting on planes parallel and perpendicular to the weld; show these stresses on a properly oriented stress element.



IN-CLASS EXAMPLE 2, cont'd

Solutions

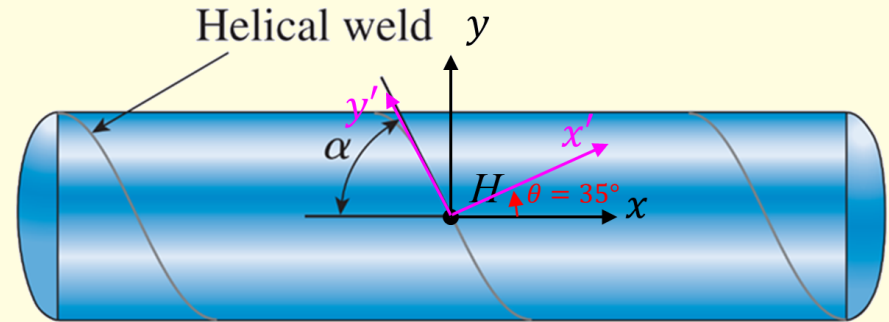
Establish xy -system as shown (z -direction is perpendicular to the surface).

(a) The circumferential (hoop) stress is

$$\sigma_{\text{hoop}} = \frac{pr}{t} = \frac{2.8 \times 10^6 \times 0.6}{0.018} = \boxed{93.333 \text{ MPa}}$$

The longitudinal stress is

$$\sigma_{\text{long}} = \frac{pr}{2t} = \frac{2.8 \times 10^6 \times 0.6}{2 \times 0.018} = \boxed{46.667 \text{ MPa}}$$



(b) The stress in the hoop and longitudinal direction is perpendicular with each other, and there is no shear stress on the point H . Therefore, the two stresses are actually the two principal stresses, i.e.

$$\sigma_1 = \sigma_{\text{hoop}} = 93.333 \text{ MPa}$$

$$\sigma_2 = \sigma_{\text{long}} = 46.667 \text{ MPa}$$

Based on the principal stresses, we can draw the Mohr's circle as shown in the figure.

Then, the maximum in-plane stress is

$$\tau_{\text{max}}^{\text{in-plane}} = R = \frac{\sigma_1 - \sigma_2}{2} = \boxed{23.333 \text{ MPa}}$$

IN-CLASS EXAMPLE 2, cont'd

Solutions, cont'd

The absolute maximum stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \boxed{46.667 \text{ MPa}}$$

(c) From the reference point $A(\sigma_x, \tau_{xy}) = A(\sigma_{\text{long}}, 0)$ to the plane along which we want to know the stresses, we need to rotate the CA in the Mohr's circle by $2\theta = 2 \times (90^\circ - \alpha) = 2 \times 35^\circ = 70^\circ$ (counterclockwise).

Using the Mohr's circle, the new stresses correspond to the point P (see figure) are

$$\sigma_{x'} = \sigma_P = \sigma_C - R \cos 2\theta = \frac{46.667 + 93.333}{2} - 23.333 \times \cos 70^\circ = \boxed{62.02 \text{ MPa}}$$

(Tensile)

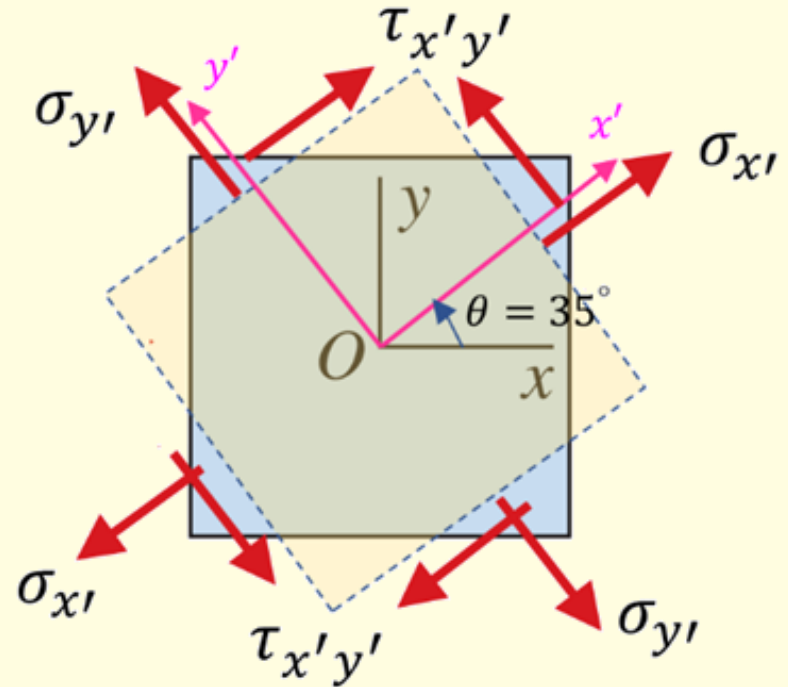
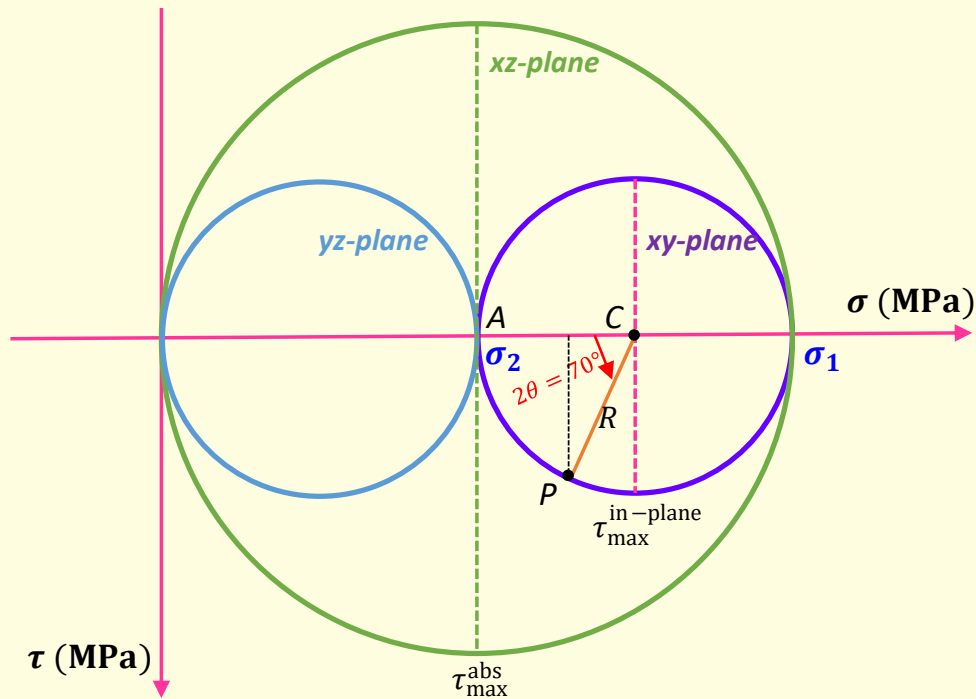
$$\tau_{x'y'} = \tau_P = R \sin 2\theta = 23.333 \times \sin 70^\circ = \boxed{21.926 \text{ MPa}}$$

$$\sigma_{y'} = \sigma_{P'} = \sigma_C + R \cos 2\theta = \boxed{77.980 \text{ MPa}} \text{ (Tensile)}$$

The oriented stress element is shown by the dashed square with stresses labeled.

IN-CLASS EXAMPLE 2, cont'd

Solutions, cont'd



RELEVANT PRACTICE

- **Assignment 4 on Blackboard, problem #5 (go to Blackboard under the “Assignments” folder, **Assignment 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**
- **Optional Extra Exercise 4 on Blackboard, problem #4 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**